UG (Sem-III) Examination-2024

(NEP : 2020)

MN-2 : PHYSICS

Time : 3 hours **Full Marks : 60**

Pass Marks : 24

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer from **both** the Groups as directed.*

Group—A

(Compulsory)

1. Answer any **four** of the following questions : $3 \times 4 = 12$

   (a) What is electric flux?

   (b) What is magnetic field?

   (c) What is electric field?

   (d) What is magnetic susceptibility?

(c) What do you mean by Quality factor?

(f) What is Bandwidth??

Group – B

(Long Answer Type Questions)

Answer any **four** questions of the following :

12×4=48

2. State and explain Conservative nature of electrostatic fields. Solve Laplace's equations for various Configurations.

3. State and explain potentials due to electric dipoles. Calculate capacitance of charged systems.

4. State and explain current loop as a magnetic dipole.

5. Derive and explain Lenz's law obeys the Law of Conservation of Energy.

6. What do you understand by electric intensity $\vec{E}$, electric polarization $\vec{P}$ and electric Displacement $\vec{D}$? Derive a relation between them.

7. State Gauss's Law in electrostatics. Derive the Differential form of this law.

8. Describe and explain mutual inductance between two circuits. Calculate the mutual inductance of two solenoids.

9. Deduce Faraday's Law of electromagnetic induction in differential form.

10. Write short notes on any **three** of the following :

$4 \times 3 = 12$

(a) B-H curves and Bandwidth

(b) AC and DC circuits

(c) Dia and Para magnetism

(d) Ballistic Galvanometer

(e) Electromagnetic damping and logarithmic damping

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